

DT-Assisted UAV Edge Network for AoI Minimization via Two-Timescale Optimization

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Abstract—

Index Terms—digital twin, age of information, XXXX

I. INTRODUCTION

Digital Twin (DT) technology is rapidly emerging as a technology that will reshape the industry. Digital twins enable real time monitoring and data driven decision making by integrating sensor data and analytics. Based on [1], employees in the oil and gas industry spend 80% of their time handling upstream data from several sources for monitoring. Creating virtual representations of field data in real time significantly enhances operational efficiency and reduces the time needed for data retrieval and validation by up to 75% leading to faster more effective decisions. Digital Twin assisted multi-access edge computing (MEC) plays an important role in ensuring real time physical to virtual twinning.

Performance Metrics XX Challenges XX AoI Motivation XXXXX Energy Motivation XXXX

II. RELATED WORK

In this section we survey selected recent relevant literature. Authors in [2] introduce a joint optimization problem to maximize digital twin data freshness reflected by the AoI metric and improve energy efficiency. They propose a light weight scheduling algorithm that compromises frequent digital twin migrations and fixed digital twin deployments. Results demonstrate up to 72.5% energy savings for a specific setup. Regulating digital twin migration across various scenarios needs further investigation. Authors in [3] propose a secure UAV-assisted digital twin framework for minimizing energy consumption and latency while ensuring secure data offloading. For the solution, authors decompose the problem and utilize successive convex approximations together with deep reinforcement learning and q-learning. Simulation results show

TABLE I
SYMBOLS USED IN FORMULATING THE PROBLEM

Symbol	Description
$\mathcal{M}$	Set of UAV edge servers
$\mathcal{E}$	Set of physical entities

a significant reduction in energy consumption with a slight reduction in latency.

III. SYSTEM MODEL

A. Network Model

We consider a digital twin (DT)-assisted UAV edge network with M UAV edge servers $\mathcal{M} = \{1, \dots, M\}$, E physical entities $\mathcal{E} = \{1, \dots, E\}$, and I IoT sensors $\mathcal{I} = \{1, \dots, I\}$.

Each entity $e \in \mathcal{E}$ is monitored by a set of sensors $\mathcal{N}_e \subseteq \mathcal{I}$. Each sensor $i \in \mathcal{I}$ belongs to exactly one entity, denoted by $e(i) \in \mathcal{E}$. We define a binary membership variable $\delta_{i,e} \in \{0, 1\}$ where $\delta_{i,e} = 1$ if sensor i belongs to entity e , and

$$\sum_{e \in \mathcal{E}} \delta_{i,e} = 1, \quad \forall i \in \mathcal{I}. \quad (1)$$

Then $\mathcal{N}_e = \{i \in \mathcal{I} : \delta_{i,e} = 1\}$.

Time is slotted with $\mathcal{T} = \{1, \dots, T\}$, each slot lasting T_s seconds. At the beginning of each slot, sensor i may generate a fresh status update indicated by $A_i(t) \in \{0, 1\}$. When $A_i(t) = 1$, the update size is $D_i(t) > 0$ bits; otherwise $D_i(t) = 0$. Sensors keep a one-packet “keep-the-latest” buffer: if a new update arrives while an old update is pending, the old one is overwritten.

Sensor i is located at $\mathbf{r}_i \in \mathbb{R}^2$. UAV m hovers at $\mathbf{q}_m \in \mathbb{R}^2$. The distances are

$$d_{i,m} = \|\mathbf{r}_i - \mathbf{q}_m\|, \quad d_{m,k} = \|\mathbf{q}_m - \mathbf{q}_k\|. \quad (2)$$

TABLE II
RELATED WORK COMPARISON

Reference Number	Aol Metric	DT	Energy	UAV(s) Deployment	UAV(s) Motion	Algorithm Used	Objective
[2]							
[3]							
[4]							
Our Work				Multiple	Stationary		

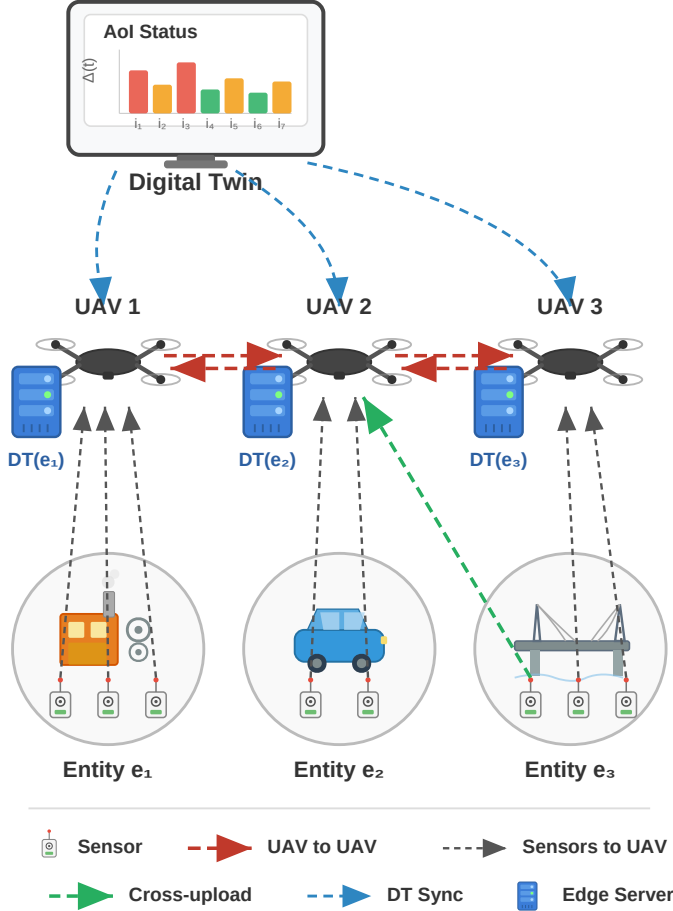


Fig. 1. System model of the DT-assisted UAV edge network with $M = 3$ UAVs, $E = 3$ entities, and $I = 8$ sensors. Dotted gray arrows show sensor-to-UAV uplink transmissions, dashed red arrows show inter-UAV backhaul links, and the dashed green arrow shows a cross-upload where a sensor of entity e_3 uploads to UAV 2 and is then forwarded to UAV 3.

Within each slot, sensors transmit via TDMA (no inter-user interference).

B. Digital Twin Placement

Each entity e must have exactly one DT replica placed on one UAV. Let $x_{e,m} \in \{0, 1\}$ indicate placement:

$$\sum_{m \in \mathcal{M}} x_{e,m} = 1, \quad \forall e \in \mathcal{E}. \quad (3)$$

DT e requires S_e bits of storage, and UAV m has capacity C_m :

$$\sum_{e \in \mathcal{E}} S_e x_{e,m} \leq C_m, \quad \forall m \in \mathcal{M}. \quad (4)$$

Let m_e be the UAV hosting entity e 's DT. For convenience, define

$$z_{i,m} \triangleq \sum_{e \in \mathcal{E}} \delta_{i,e} x_{e,m} \in \{0, 1\}, \quad (5)$$

which indicates whether sensor i 's DT is on UAV m .

C. Bandwidth Split, Scheduling, and Uplink Transmission

The total bandwidth B_{tot} is split into access (sensor-to-UAV) and backhaul (UAV-to-UAV):

$$B^{\text{ac}} + B^{\text{bh}} \leq B_{\text{tot}}, \quad B^{\text{ac}} \geq 0, \quad B^{\text{bh}} \geq 0. \quad (6)$$

Buffer state. For each sensor, $Q_i(t) \in \{0, 1\}$ indicates whether there is a pending update, $W_i(t) \geq 0$ is its size, and $U_i(t) \geq 0$ is the waiting time (in slots). These three quantities are updated jointly at the start of each slot, accounting for both the transmission outcome of the previous slot and any new arrival:

$$Q_i(t) = \max\{Q_i(t-1)(1 - c_i(t-1)), A_i(t)\}, \quad (7)$$

$$W_i(t) = A_i(t) D_i(t) + (1 - A_i(t)) W_i(t-1)(1 - c_i(t-1)), \quad (8)$$

$$U_i(t) = \begin{cases} 0, & A_i(t) = 1, \\ (U_i(t-1) + 1)(1 - c_i(t-1)), & A_i(t) = 0, Q_i(t-1) = 1, \\ 0, & A_i(t) = 0, Q_i(t-1) = 0, \end{cases} \quad (9)$$

where $c_i(t-1) = \sum_m a_{i,m}(t-1) \in \{0, 1\}$ indicates whether sensor i was served in the previous slot. Equations (7)–(9) capture both the keep-the-latest overwrite rule (a new arrival replaces the old packet) and the post-transmission clearing (a successfully transmitted packet is removed), without requiring a separate buffer-reset equation.

TDMA scheduling. Let $a_{i,m}(t) \in \{0,1\}$ indicate whether sensor i uploads to UAV m in slot t :

$$\sum_{m \in \mathcal{M}} a_{i,m}(t) \leq Q_i(t), \quad \forall i, t, \quad (10)$$

$$\sum_{i \in \mathcal{I}} a_{i,m}(t) \leq 1, \quad \forall m, t. \quad (11)$$

Uplink rate. With channel gain $h_{i,m} = \varphi(d_{i,m})$, transmit power $p_{i,m}(t)$, and noise PSD N_0 :

$$R_{i,m}^{\text{ul}}(t) = B^{\text{ac}} \log_2 \left(1 + \frac{p_{i,m}(t) h_{i,m}}{N_0 B^{\text{ac}}} \right). \quad (12)$$

The uplink transmission time is $\tau_{i,m}^{\text{ul}}(t) = W_i(t)/R_{i,m}^{\text{ul}}(t)$. We assume per-sensor power constraints:

$$0 \leq p_{i,m}(t) \leq p_i^{\text{max}}, \quad \forall i, m, t. \quad (13)$$

D. Backhaul Forwarding

If sensor i uploads to UAV m but its DT is on another UAV ($m_{e(i)} \neq m$), the data is forwarded on the backhaul.

Backhaul bandwidth B^{bh} is split across directed links by shares $b_{m,k} \geq 0$:

$$\sum_{m \in \mathcal{M}} \sum_{\substack{k \in \mathcal{M} \\ k \neq m}} b_{m,k} \leq B^{\text{bh}}. \quad (14)$$

The backhaul rate between UAV m and UAV k is

$$R_{m,k}^{\text{bh}}(t) = b_{m,k} \log_2 \left(1 + \frac{p_{m,k}(t) h_{m,k}}{N_0 b_{m,k}} \right), \quad (15)$$

where $h_{m,k} = \varphi(d_{m,k})$ and $p_{m,k}(t)$ is the backhaul transmit power. The forwarding time for sensor i is $\tau_{i,m \rightarrow m_{e(i)}}^{\text{bh}}(t) = W_i(t)/R_{m,m_{e(i)}}^{\text{bh}}(t)$. Backhaul power is bounded by

$$0 \leq p_{m,k}(t) \leq p^{\text{maxbh}}_{m,k}, \quad \forall m, k, t. \quad (16)$$

E. Processing at the DT Host (Constant CPU)

Once the update arrives at the hosting UAV, the DT processes it. Entity e requires κ_e CPU cycles per bit. To keep the model simple, we assume a constant CPU rate $f > 0$ that is used for all entities and does not change with time. The processing time for sensor i 's update is

$$\tau_i^{\text{proc}}(t) = \frac{\kappa_{e(i)} W_i(t)}{f}. \quad (17)$$

If UAV m hosts $n_m = \sum_{e \in \mathcal{E}} x_{e,m}$ DTs, we enforce the CPU budget by

$$f \sum_{e \in \mathcal{E}} x_{e,m} \leq f_m^{\text{max}}, \quad \forall m. \quad (18)$$

F. End-to-End Delay and AoI

Let $c_i(t) \triangleq \sum_m a_{i,m}(t)$ indicate whether sensor i is served in slot t . The end-to-end delay includes uplink, backhaul when needed, and processing:

$$d_i(t) = \sum_{m \in \mathcal{M}} a_{i,m}(t) \left[\tau_{i,m}^{\text{ul}}(t) + (1 - z_{i,m}) \tau_{i,m \rightarrow m_{e(i)}}^{\text{bh}}(t) + \tau_i^{\text{proc}}(t) \right]. \quad (19)$$

We use the normalized delay $\hat{d}_i(t) = d_i(t)/T_s$. The slot duration T_s defines the decision interval; delays may exceed one slot, and this is captured through $\hat{d}_i(t)$.

The AoI for sensor i at its DT is

$$\Delta_i(t+1) = (1 - c_i(t))(\Delta_i(t) + 1) + c_i(t)(U_i(t) + \hat{d}_i(t)), \quad (20)$$

with $\Delta_i(1) = 0$. Entity-level AoI is defined as the maximum AoI among its sensors:

$$\Delta_e(t) = \max_{i \in \mathcal{N}_e} \Delta_i(t). \quad (21)$$

IV. PROBLEM FORMULATION

A. Energy Model

The uplink energy spent by sensor i is

$$E_i^{\text{ul}}(t) = \sum_{m \in \mathcal{M}} a_{i,m}(t) p_{i,m}(t) \tau_{i,m}^{\text{ul}}(t). \quad (22)$$

For the UAV backhaul, we use a power-time model:

$$E_m^{\text{bh}}(t) = \sum_{\substack{k \in \mathcal{M} \\ k \neq m}} p_{m,k}(t) \tau_{m,k}^{\text{bh}}(t), \quad (23)$$

where $\tau_{m,k}^{\text{bh}}(t)$ is the total backhaul transmission time scheduled on link (m,k) during slot t . Since each UAV serves at most one sensor per slot, at most one forwarding transmission occurs per UAV per slot, hence

$$\tau_{m,k}^{\text{bh}}(t) = \sum_{i \in \mathcal{I}} a_{i,m}(t) (1 - z_{i,m}) \mathbf{1}\{m_{e(i)} = k\} \frac{W_i(t)}{R_{m,k}^{\text{bh}}(t)}. \quad (24)$$

We enforce a long-term UAV energy budget:

$$\frac{1}{T} \sum_{t=1}^T E_m^{\text{bh}}(t) \leq E_m^{\text{max}}, \quad \forall m. \quad (25)$$

B. Objective

We minimize the average entity AoI:

$$\bar{\Delta} \triangleq \frac{1}{T} \sum_{t=1}^T \frac{1}{|\mathcal{E}|} \sum_{e \in \mathcal{E}} \Delta_e(t). \quad (26)$$

C. Optimization Problem

$$\min_{\{a_{i,m}(t)\}, \{x_{e,m}\}, \{p_{i,m}(t)\}, \{\mathbf{q}_m\}, \{p_{m,k}(t)\}, \{b_{m,k}\}, \{B^{\text{ac}}, B^{\text{bh}}\}} \bar{\Delta} \quad (27a)$$

$$\text{s.t.} \quad (27b)$$

$$x_{e,m} \in \{0, 1\}, \sum_m x_{e,m} = 1, \forall e, \quad (27c)$$

$$\sum_e S_e x_{e,m} \leq C_m, \forall m, \quad (27d)$$

$$a_{i,m}(t) \in \{0, 1\}, \sum_m a_{i,m}(t) \leq Q_i(t), \forall i, t, \quad (27e)$$

$$\sum_i a_{i,m}(t) \leq 1, \forall m, t, \quad (27f)$$

$$B^{\text{ac}} \geq 0, B^{\text{bh}} \geq 0, B^{\text{ac}} + B^{\text{bh}} \leq B_{\text{tot}}, \quad (27g)$$

$$b_{m,k} \geq 0, \sum_m \sum_{k \neq m} b_{m,k} \leq B^{\text{bh}}, \quad (27h)$$

$$0 \leq p_{i,m}(t) \leq p_i^{\text{max}}, \forall i, m, t, \quad (27i)$$

$$0 \leq p_{m,k}(t) \leq p_{m,k}^{\text{maxbh}}, \forall m, k, t, \quad (27j)$$

$$f \sum_e x_{e,m} \leq f_m^{\text{max}}, \forall m, \quad (27k)$$

$$(7)-(9) \text{ hold } \forall i, t, \quad (27l)$$

$$\Delta_i(t+1) = (1 - c_i(t))(\Delta_i(t) + 1) \forall i, t, \quad (27m)$$

$$+ c_i(t)(U_i(t) + \hat{d}_i(t)), \quad (27n)$$

$$\Delta_e(t) = \max_{i \in \mathcal{N}_e} \Delta_i(t), \Delta_e(t) \leq \Delta^{\text{max}}, \forall e, t, \quad (27o)$$

$$\frac{1}{T} \sum_{t=1}^T E_m^{\text{bh}}(t) \leq E_m^{\text{max}}, \forall m. \quad (27p)$$

where $c_i(t) = \sum_m a_{i,m}(t)$ and the buffer evolves according to (7)–(9).

The above problem is challenging because it mixes binary placement and scheduling decisions with nonlinear transmission rates, and the AoI recursion couples decisions across time. In addition, DT placement impacts whether forwarding is required, which changes the end-to-end delay and thus affects the AoI.

V. PROPOSED SOLUTION

We solve the problem with a two-timescale reinforcement learning approach called **Lya-HPPO** (Lyapunov-driven Hierarchical Proximal Policy Optimization). The idea is to separate the slow structural decisions from the fast slot-level decisions, and to handle the long-term power constraint using a Lyapunov virtual queue instead of a learned penalty weight. This design follows the framework introduced in [5] for related UAV-AoI problems, which itself builds on the Lyapunov-DRL combination originally proposed in [6] and the stochastic network optimization theory of [7].

A. Two-Timescale Decomposition

The structural decisions, which are the UAV positions $\mathbf{q}_m$ and the DT placement $x_{e,m}$, are taken on a slow timescale: once every H slots. The slot-level decisions, which are the

TDMA scheduling $a_{i,m}(t)$, the uplink power $p_{i,m}(t)$, and the backhaul power $p_{m,k}(t)$, are taken on the fast timescale: every slot. We use two separate policies:

- A **manager** agent, which selects $(\mathbf{q}_m, x_{e,m})$ at the beginning of every H -slot window.
- A **worker** agent, which selects the slot-level actions at every slot.

Both agents are trained with Proximal Policy Optimization (PPO).

B. Handling the Long-Term Power Constraint

The objective is the average AoI in (26), but the system is also subject to a long-term constraint on the average uplink power

$$\mathbb{E}[\bar{P}_{\text{ul}}] \leq P_{\text{budget}}. \quad (28)$$

This kind of constraint is hard for standard reinforcement learning because the constraint is defined on a long-term average, while the policy is updated from short rollouts. To deal with this, we introduce a virtual queue $Z(k)$ at the slow timescale.

C. The Virtual Queue Z

The virtual queue $Z(k)$ is a single non-negative number maintained per episode. It is updated after each manager step using

$$Z(k+1) = \max\{Z(k) + (\bar{P}_{\text{ul}}(k) - P_{\text{budget}}), 0\}, \quad (29)$$

where $\bar{P}_{\text{ul}}(k)$ is the average uplink power measured during the H fast slots of manager step k . The queue grows when the policy goes over budget and shrinks (down to zero) when it stays under budget. The $\max\{\cdot, 0\}$ keeps the queue non-negative so that the policy cannot “store credit” for past good behavior.

The value of $Z(k)$ is used in two places:

- It is part of the manager’s observation, so the policy can see how badly it has been violating the budget recently.
- It scales the power-violation term in the manager’s reward, so the penalty for going over budget grows automatically when violations accumulate.

This is the standard Lyapunov drift-plus-penalty mechanism. It is shown in [7] that the stability of $Z(k)$ is equivalent to the long-term constraint (28) being satisfied.

D. Manager and Worker Rewards

The manager reward at slow step k is

$$r^{\text{m}}(k) = -V \cdot \bar{\Delta}(k) - Z(k) \cdot (\bar{P}_{\text{ul}}(k) - P_{\text{budget}}) - \lambda_{\text{bh}} \cdot v_{\text{bh}}(k), \quad (30)$$

where V is a fixed positive weight that controls the trade-off between AoI and constraint satisfaction, and $v_{\text{bh}}(k)$ captures the backhaul-budget violation. We use a fixed value of V rather than a learned dual variable; this is the main difference from PPO-Lagrangian methods and is what makes the training stable. According to [7], the time-averaged AoI converges to within $O(1/V)$ of the optimal feasible value, with the average queue length growing as $O(V)$.

The worker reward is simpler and contains AoI, a small fixed soft penalty for uplink power, and a soft penalty for backhaul violation. The worker does not see $Z(k)$. The constraint is enforced through the manager.

E. Training Procedure

We train the two agents in two stages:

- 1) **Worker training.** The worker is trained first using PPO. The deployment $(\mathbf{q}_m, x_{e,m})$ is randomized at the start of each episode so that the worker learns a robust policy that handles any deployment.
- 2) **Manager training.** The worker is then frozen and the manager is trained using PPO on a slow-timescale environment. The state of this environment includes the UAV positions, DT placement, average AoI, average uplink and backhaul power, and the virtual queue $Z(k)$. The reward is given by (30). The queue $Z(k)$ is updated automatically inside the environment using (29).

This staged approach decouples the two timescales and avoids the joint-training instability that we observed in earlier experiments with multi-agent and constrained variants of PPO.

F. Baselines

We compare the trained Lya-HPPO manager against three deployment baselines that share the same trained worker:

- **K-means:** UAVs are placed at the centroids of sensor clusters, and each DT is hosted on the closest UAV.
- **AoI-greedy:** UAVs are placed at the centroids of the entities with the largest current AoI.
- **Random:** UAV positions and DT placement are sampled uniformly.

This isolates the contribution of the deployment policy, since the slot-level controller is identical across the four methods.

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